AWS CLOUD

Launch one **EC2 instance** using **Amazon Linux 2**.

Install **Docker**.

Steps to install docker:

1. **ssh -i your-key.pem ec2-user@your-public-ip** used this connected Ip address in CD
2. **sudo yum update -y** to update the docker
3. we can verify the version with **docker –version**

```
[root@ip-172-31-18-24 ~]# sudo yum install docker -y
Amazon Linux 2023 Kernel Livepatch repository                    252 kB/s |  31 kB     00:00
Dependencies resolved.
==================================================================================================
 Package                      Architecture   Version                      Repository        Size
==================================================================================================
Installing:
 docker                       x86_64         25.0.14-1.amzn2023.0.2       amazonlinux       46 M
Installing dependencies:
 container-selinux            noarch         4:2.245.0-1.amzn2023         amazonlinux       58 k
 containerd                   x86_64         2.2.1-1.amzn2023.0.1         amazonlinux       24 M
 iptables-libs                x86_64         1.8.8-3.amzn2023.0.2         amazonlinux      401 k
 iptables-nft                 x86_64         1.8.8-3.amzn2023.0.2         amazonlinux      183 k
 libcgroup                    x86_64         3.0-1.amzn2023.0.1           amazonlinux       75 k
 libnetfilter_conntrack       x86_64         1.0.8-2.amzn2023.0.2         amazonlinux       58 k
 libnfnetlink                 x86_64         1.0.1-19.amzn2023.0.2        amazonlinux       30 k
 libnftnl                     x86_64         1.2.2-2.amzn2023.0.2         amazonlinux       84 k
 pigz                         x86_64         2.5-1.amzn2023.0.3           amazonlinux       83 k
 runc                         x86_64         1.3.4-1.amzn2023.0.2         amazonlinux      3.9 M

Transaction Summary
```

```
omplete!
[root@ip-172-31-18-24 ~]# sudo yum update -y
ast metadata expiration check: 0:01:59 ago on Mon Apr 13 09:39:02 2026.
ependencies resolved.
othing to do.
omplete!
[root@ip-172-31-18-24 ~]#
```

4. for installation and check the server where running and

sudo yum install docker -y

sudo **systemctl** start docker

sudo **systemctl** enable docker

```
[root@ip-172-31-18-24 ~]# docker --version
Docker version 25.0.14, build 0bab007
[root@ip-172-31-18-24 ~]#
```

```
[root@ip-172-31-18-24 ~]# sudo systemctl start docker
[root@ip-172-31-18-24 ~]# sudo systemctl enable docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /
vice.
[root@ip-172-31-18-24 ~]# sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset:
     Active: active (running) since Mon 2026-04-13 09:45:04 UTC; 34s ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 27427 (dockerd)
      Tasks: 9
     Memory: 30.4M
        CPU: 342ms
     CGroup: /system.slice/docker.service
             └─27427 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/co
```

Install **Jenkins**.

Steps to install Jenkins:

1. we can go the Jenkins website to download from there we can get the link we can it paste and download the Jenkins of repository.

sudo wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/redhat-stable/jenkins.repo

```
[root@ip-172-31-18-24 ~]# sudo wget -O /etc/yum.repos.d/jenkins.repo \
    https://pkg.jenkins.io/rpm-stable/jenkins.repo
--2026-04-13 09:59:39--  https://pkg.jenkins.io/rpm-stable/jenkins.repo
Resolving pkg.jenkins.io (pkg.jenkins.io)... 146.75.38.133, 2a04:4e42:78::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|146.75.38.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 267 [application/octet-stream]
Saving to: '/etc/yum.repos.d/jenkins.repo'

/etc/yum.repos.d/jenkins.repo   100%[=================================================>]     267  --.-KB/s    in 0s
```

2. We can install the latest **java version**

sudo yum install **java-11-amazon-corretto -y**

```
Complete!
[root@ip-172-31-18-24 ~]# sudo yum install java-11-amazon-corretto -y
Last metadata expiration check: 0:00:24 ago on Mon Apr 13 09:59:59 2026.
Dependencies resolved.
==========================================================================================
 Package                    Architecture    Version                      Repository    Size
==========================================================================================
Installing:
 java-11-amazon-corretto    x86_64          1:11.0.30+7-1.amzn2023       amazonlinux   197 k
```

```
Complete!
[root@ip-172-31-18-24 ~]# java -version
openjdk version "11.0.30" 2026-01-20 LTS
OpenJDK Runtime Environment Corretto-11.0.30.7.1 (build 11.0.30+7-LTS)
OpenJDK 64-Bit Server VM Corretto-11.0.30.7.1 (build 11.0.30+7-LTS, mixed mode)
[root@ip-172-31-18-24 ~]#
```

3. To check the version: **java -version**

4.After that we can install Jenkins by using this command

sudo yum install jenkins -y

```
OpenJDK 64-Bit Server VM Corretto-11.0.30.7.1 (build 11.0.30+7-LTS, mixed mode)
[root@ip-172-31-18-24 ~]# sudo yum install jenkins -y
Last metadata expiration check: 0:04:45 ago on Mon Apr 13 09:59:59 2026.
Dependencies resolved.
==========================================================================================
 Package          Architecture        Version            Repository          Size
==========================================================================================
Installing:
 jenkins          noarch              2.541.3-1          jenkins             92 M

Transaction Summary
==========================================================================================
Install  1 Package
```

5. we can check the installation done and it is running or not by using these commands

sudo **systemctl start** jenkins

sudo **systemctl enable** Jenkins

sudo **systemctl status** Jenkins

```
[root@ip-172-31-18-24 ~]# sudo systemctlstatus jenkins
sudo: systemctlstatus: command not found
[root@ip-172-31-18-24 ~]# sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: disabled)
     Active: active (running) since Mon 2026-04-13 10:16:57 UTC; 31s ago
   Main PID: 29763 (java)
      Tasks: 42 (limit: 1014)
     Memory: 264.0M
        CPU: 17.985s
     CGroup: /system.slice/jenkins.service
             └─29763 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/je

Apr 13 10:16:53 ip-172-31-18-24.ec2.internal jenkins[29763]: [LF]>
Apr 13 10:16:53 ip-172-31-18-24.ec2.internal jenkins[29763]: [LF]> This may also be found at: /var/lib/jenkins/secre
Apr 13 10:16:53 ip-172-31-18-24.ec2.internal jenkins[29763]: [LF]>
Apr 13 10:16:53 ip-172-31-18-24.ec2.internal jenkins[29763]: [LF]> ****************************************************
Apr 13 10:16:53 ip-172-31-18-24.ec2.internal jenkins[29763]: [LF]> ****************************************************
Apr 13 10:16:53 ip-172-31-18-24.ec2.internal jenkins[29763]: [LF]> ****************************************************
Apr 13 10:16:57 ip-172-31-18-24.ec2.internal jenkins[29763]: 2026-04-13 10:16:57.516+0000 [id=30]        INFO
Apr 13 10:16:57 ip-172-31-18-24.ec2.internal jenkins[29763]: 2026-04-13 10:16:57.539+0000 [id=23]        INFO
Apr 13 10:16:57 ip-172-31-18-24.ec2.internal systemd[1]: Started jenkins.service - Jenkins Continuous Integration Se
Apr 13 10:17:02 ip-172-31-18-24.ec2.internal jenkins[29763]: 2026-04-13 10:17:02.615+0000 [id=61]        WARNING
lines 1-20/20 (END)
```

Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:

`/var/lib/jenkins/secrets/initialAdminPassword`

Please copy the password from either location and paste it below.

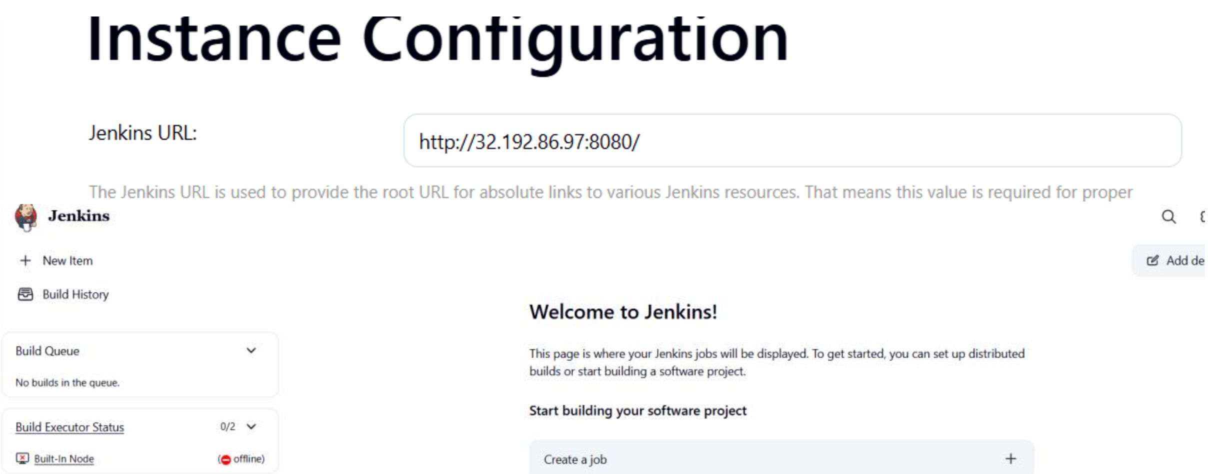
Administrator password

Continue

```
[root@ip-172-31-18-24 ~]# cat /var/lib/jenkins/secrets/initialAdminPassword
c763654c3dcc42a0b77a2c3806ead1e1
[root@ip-172-31-18-24 ~]# ^C
[root@ip-172-31-18-24 ~]#
```

Getting Started

Folders	✓ OWASP Markup Formatter	✓ Build Timeout	Credentials Binding	** commons-lang3 v3.x Jenkins API ** Ionicons API Folders OWASP Markup Formatter ** ASM API ** JSON Path API ** Structs ** Pipeline: Step API ** commons-text API ** Token Macro Build Timeout ** bouncycastle API
Timestamper	Workspace Cleanup	Ant	Gradle	
Pipeline	GitHub Branch Source	Pipeline: GitHub Groovy Libraries	Pipeline Graph View	
Git	SSH Build Agents	Matrix Authorization Strategy	LDAP	
Email Extension	Mailer	Dark Theme		



Install **Apache.** (httpd)

Steps to install Httpd:

1.We can install Httpd by using these commands

sudo yum **install httpd -y**

sudo **systemctl start** httpd

```
[root@ip-172-31-18-24 ~]# sudo yum install httpd -y
Last metadata expiration check: 0:47:51 ago on Mon Apr 13 09:59:59 2026.
Dependencies resolved.
================================================================================================
 Package            Architecture     Version                        Repository          Size
================================================================================================
Installing:
 httpd              x86_64           2.4.66-1.amzn2023.0.1          amazonlinux         47 k
Installing dependencies:
 apr                x86_64           1.7.5-1.amzn2023.0.4           amazonlinux        129 k
 apr-util           x86_64           1.6.3-1.amzn2023.0.2           amazonlinux         97 k
 apr-util-lmdb      x86_64           1.6.3-1.amzn2023.0.2           amazonlinux         13 k
```

sudo **systemctl enable** httpd

```
[root@ip-172-31-18-24 ~]# sudo systemctl start httpd
[root@ip-172-31-18-24 ~]# sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/ht
[root@ip-172-31-18-24 ~]# sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
     Active: active (running) since Mon 2026-04-13 10:48:14 UTC; 2min 59s ago
       Docs: man:httpd.service(8)
   Main PID: 31492 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 1014)
     Memory: 13.9M
```

We can go to website and check with the port number with IP address we can get page as if it works

Install **Nginx.**

Steps to install Nginx:

1. We can install the Nginx by using **these commands**

sudo apt install nginx

```
[root@ip-172-31-18-24 ~]# sudo apt install nginx
sudo: apt: command not found
[root@ip-172-31-18-24 ~]# sudo yum install nginx
Last metadata expiration check: 0:59:18 ago on Mon Apr 13 09:59:59 2026.
Dependencies resolved.
==========================================================================================================
 Package                  Architecture       Version                               Repository       Size
==========================================================================================================
Installing:
 nginx                    x86_64             1:1.28.2-1.amzn2023.0.1               amazonlinux      33 k
Installing dependencies:
```

2. once it install, we can use these commands to check it is running or not

sudo **systemctl start** nginx

sudo **systemctl enable** nginx

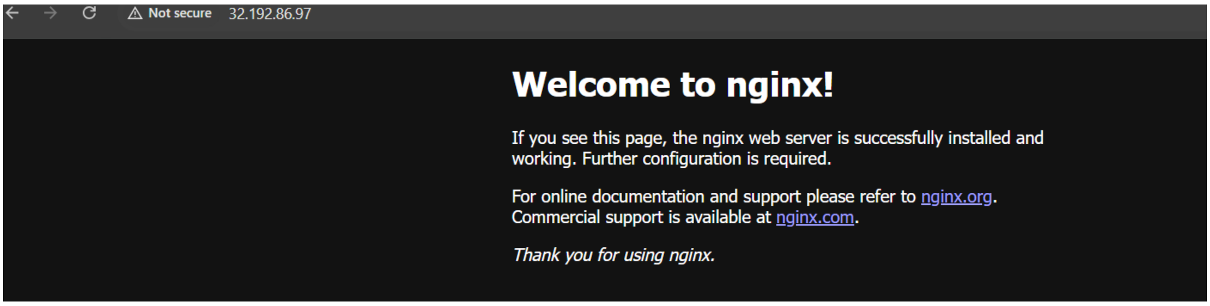
sudo **systemctl status** nginx

```
[root@ip-172-31-18-24 ~]# sudo systemctl start nginx
[root@ip-172-31-18-24 ~]# sudo systemctl enable nginx
[root@ip-172-31-18-24 ~]# sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
     Active: active (running) since Mon 2026-04-13 11:01:53 UTC; 8s ago
   Main PID: 33270 (nginx)
      Tasks: 3 (limit: 1014)
     Memory: 3.4M
        CPU: 58ms
     CGroup: /system.slice/nginx.service
             ├─33270 "nginx: master process /usr/sbin/nginx"
             ├─33271 "nginx: worker process"
             └─33272 "nginx: worker process"
```

3. We can get **it is running** its work and then go to the new website and check the port no

4. Previously we changed the **port no: Httpd** same way we must change here with **port 81**

5. We can get the follow welcome page of Welcome to Nginx it is working.



Install **Apache Tomcat.**

Steps to install Apache Tomcat:

1.To install Tomcat we can go to the website and type tomcat download from there we can get
the tar version and copy the link and paste in cd.

sudo wget https://downloads.apache.org/tomcat/tomcat-9/v9.0.85/bin/apache-tomcat-9.0.85.tar.gz

2.We can extract the tomcat by using tar command

```
[root@ip-172-31-18-24 ~]# sudo wget https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.117/bin/apache-tomcat-9.0.117.tar.gz
--2026-04-13 11:08:42--  https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.117/bin/apache-tomcat-9.0.117.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13089813 (12M) [application/x-gzip]
Saving to: 'apache-tomcat-9.0.117.tar.gz'

apache-tomcat-9.0.117.tar.gz    100%[==================================================>]  12.48M  --.-KB/s    in 0.05s

2026-04-13 11:08:42 (244 MB/s) - 'apache-tomcat-9.0.117.tar.gz' saved [13089813/13089813]

[root@ip-172-31-18-24 ~]#
```

sudo tar -xvzf apache-tomcat-9.0.85.tar.gz and move to tar to tomcat file name changed

sudo mv apache-tomcat-9.0.85 tomcat

```
[root@ip-172-31-18-24 ~]# ls
apache-tomcat-9.0.117.tar.gz
[root@ip-172-31-18-24 ~]# tar xvf apache-tomcat-9.0.117.tar.gz
apache-tomcat-9.0.117/conf/
apache-tomcat-9.0.117/conf/catalina.policy
```

3.From there we can tomcat installation and change the owner name and start the start startup shell by using the below commands.

cd /tomcat/bin

```
[root@ip-172-31-18-24 ~]# cd tomcat
[root@ip-172-31-18-24 tomcat]# cd bin
[root@ip-172-31-18-24 bin]# ls
bootstrap.jar        ciphers.sh                    daemon.sh      setclasspath.bat  startup.sh
catalina-tasks.xml   commons-daemon-native.tar.gz  digest.bat     setclasspath.sh   tomcat-jul
catalina.bat         commons-daemon.jar            digest.sh      shutdown.bat      tomcat-nat
catalina.sh          configtest.bat                makebase.bat   shutdown.sh       tool-wrapp
ciphers.bat          configtest.sh                 makebase.sh    startup.bat       tool-wrapp
[root@ip-172-31-18-24 bin]# sh startup.sh
Using CATALINA_BASE:   /root/tomcat
Using CATALINA_HOME:   /root/tomcat
Using CATALINA_TMPDIR: /root/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /root/tomcat/bin/bootstrap.jar:/root/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
```

sudo chmod +x *.sh

sudo ./startup.sh

4. We have to change the port no from **8080 to 8082**

```
    -->
    <Connector port="8082" protocol="HTTP/1.1"
               connectionTimeout="20000"
               redirectPort="8443"
               maxParameterCount="1000"
               />
    <!-- A "Connector" using the shared thread pool-->
```

5. Below is the tomcat port no we can change in server default it comes we have to change and save and exit.

sudo vi /opt/tomcat/conf/server.xml

6.Then again, we have to restart it and check it is running by below commands

cd /opt/tomcat/bin

sudo ./shutdown.sh

sudo ./startup.sh

```
[root@ip-172-31-18-24 bin]# sh startup.sh
Using CATALINA_BASE:   /root/tomcat
Using CATALINA_HOME:   /root/tomcat
Using CATALINA_TMPDIR: /root/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /root/tomcat/bin/bootstrap.jar:/root/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
```